Qualifying Exam

November 7, 2025

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Motivation

Secure computation has taken great leaps in recent years...

but it's still impractical in many scenarios.

Goal: Fill in the gaps that prevent the use of MPC in practice

Scaling Up MPC

1. More sophisticated functions

- Oblivious shuffling
- Improving the expressivity of MPC

2. Directly incorporating more parties

- Significant practical challenges
- Distribute trust more widely

The Vision

Efficient secure computation involving millions of parties for complex functions.

Part 1: For a fixed system model, improve the efficiency of a complex building block.

Part 2: For a moderately complex workload, scale the number of parties.

Roadmap

1. Oblivious Shuffle Preprocessing

2. Scaling with Many Parties

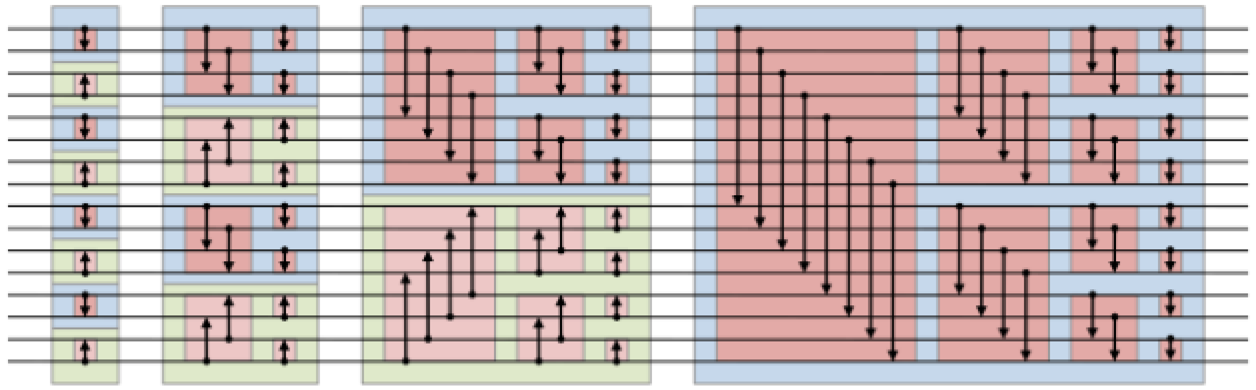
Oblivious Shuffle Preprocessing

Oblivious Shuffling

Two parties, P_0 and P_1, want to shuffle a secret-shared vector $[\mathbf{x}]$ by a permutation π.

- Neither party should learn anything about π
- The output is a secret sharing $[\pi(\mathbf{x})]$

Example: Oblivious Sorting



Oblivious sorting networks require $O(n \log^2 n)$ comparison gates.

- We want to match the plaintext $O(n \log n)$ algorithms

The shuffle-then-sort paradigm (Hamada et al.)

- Obliviously shuffle, then use a data-dependent sorting algorithm
- The revealed permutation is uncorrelated with the input
- Achieves $O(n \log n)$ secure comparisons

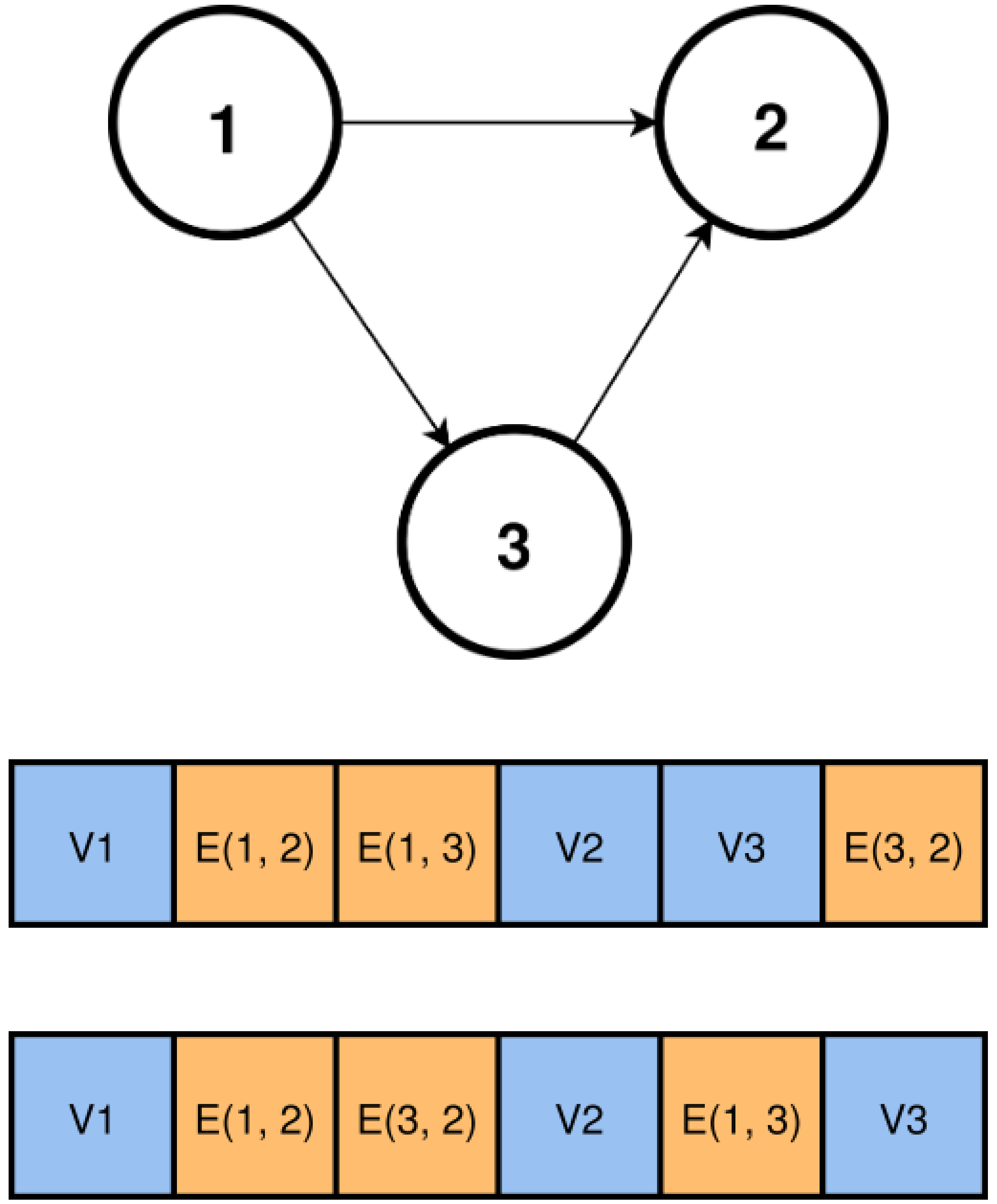
Example: Graph Algorithms

Not easy to efficiently express a graph obliviously.

- Lots of data-dependent operations

The GraphSC paradigm (Nayak et al.)

- Encode the graph as a single list
- The list has two orderings
- Two important operations, scatter and gather
- Each operation is efficient only in one of the orderings
- Shuffling allows us to convert between the orderings



Broader Applications

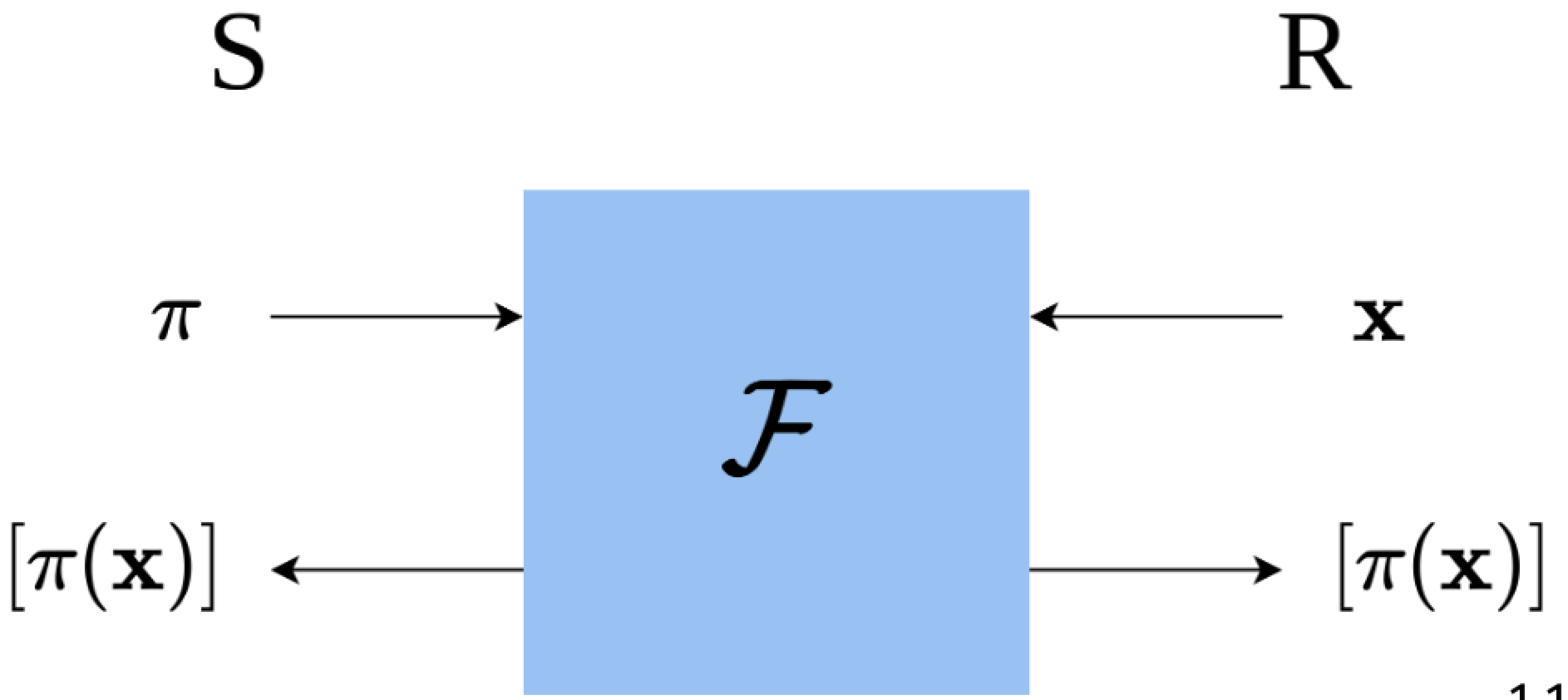
Combine aspects of the circuit model and RAM model of computation.

- A bridge between the two worlds
- Expressivity makes it central to complex workloads

How do we do it?

Break the Problem in Half

- Let $\pi = \pi_1 \circ \pi_0$ be a composition of random permutations
  - P_0 knows π_0
  - P_1 knows π_1
  - Neither party knows anything about π
- One party acts as a sender, the other as a receiver
- Known as "Permute+Share" (Chase et al.)



Preprocessing

Permute+Share can be broken into preprocessing and online phases.

- The preprocessing phase shuffles pseudorandom values
- The online phase derandomizes in a single message

Preprocessing produces a *permutation correlation*.

$$(\pi, C), (A, B)$$

$$\pi(A) = B + C$$

Online Derandomization Protocol

S: (π, C) R: $(\mathbf{x}, (A, B))$

$$\xleftarrow{\quad\quad\mathbf{x} - A\quad\quad}$$

$\pi(\mathbf{x} - A) + C$ B

$$\begin{aligned}
\text{output} &= \pi(\mathbf{x} - A) + C + B \\
&= \pi(\mathbf{x} - A) + \pi(A) \\
&= \pi(\mathbf{x}) - \pi(A) + \pi(A) \\
&= \pi(\mathbf{x})
\end{aligned}$$

Formally Defining the Problem

Functionality $\mathcal{F}_{\text{GenPerm}}$

PUBLIC PARAMETERS: Number of elements n and bitwidth ℓ.
INPUT: None
OUTPUT: The functionality samples a uniformly random permutation $\pi \leftarrow S_n$ and uniformly random vectors $A, B, C \in \{0,1\}^{n \times \ell}$, such that $\pi(A) = B + C$. $\mathcal{F}_{\text{GenPerm}}$ outputs (π, C) to the sender and (A, B) to the receiver.

Three parameters we care about:

- n, the number of elements in a permutation correlation
- ℓ, the bitwidth of each element of A, B, and C in a single permutation correlation
- p, the total number of permutation correlations generated

State of the Art

Peceny, Raghuraman, Rindal, and Shah (PKC 2025)

- Two constructions
- Both based on oblivious pseudorandom functions

Construction 1	Construction 2
$O(n\ell)$	$O(n \log \ell)$

Efficient Permutation Correlations and Batched Random Access for Two-Party Computation

Stanislav Peceny[1*], Srinivasan Raghuraman[2], Peter Rindal[3], and Harshal Shah[3]

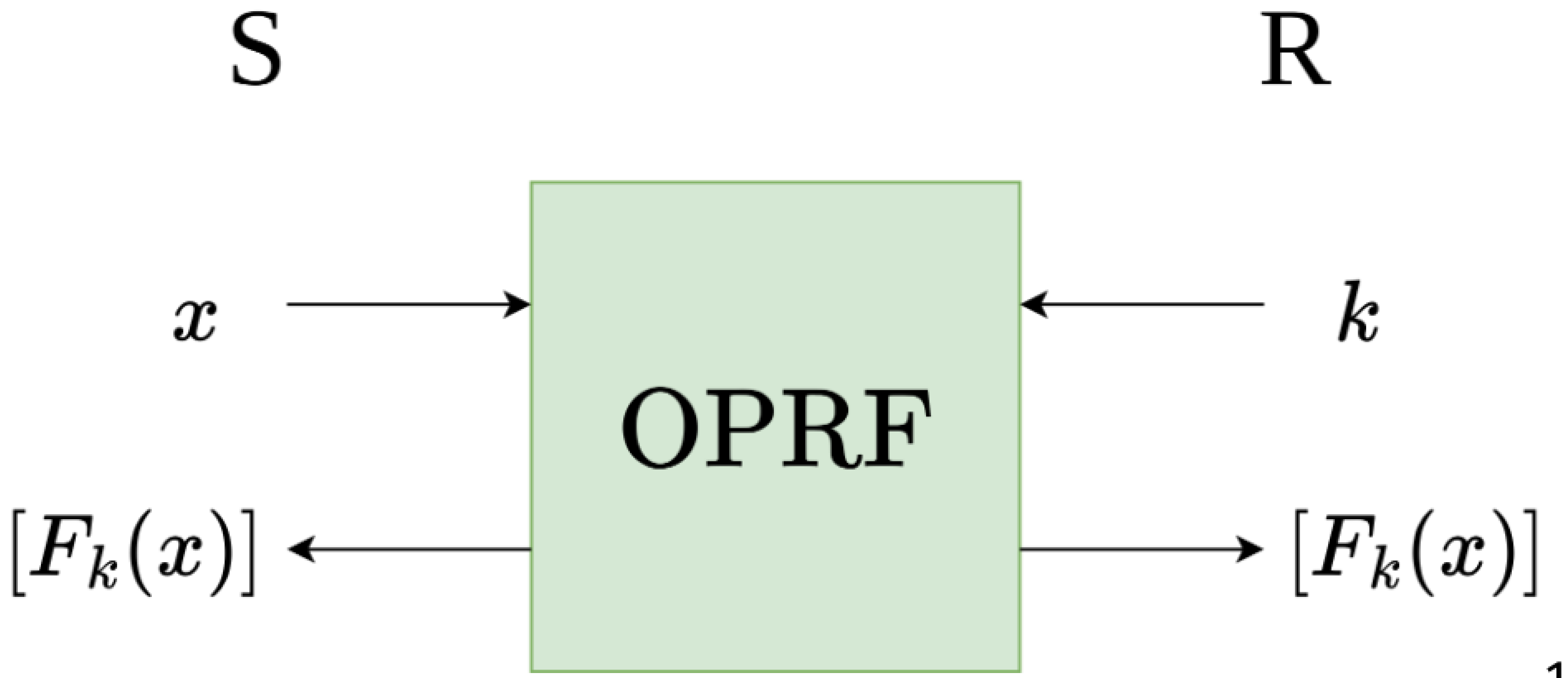
[1] Georgia Institute of Technology
[2] Visa Research and MIT
[3] Visa Research

Abstract. In this work we formalize the notion of a two-party permutation correlation $(A, B), (C, \pi)$ s.t. $\pi(A) = B + C$ for a random permutation π of n elements and vectors $A, B, C \in \mathbb{F}^n$. This correlation can be viewed as an abstraction and generalization of the Chase et al. (Asiacrypt 2020) share translation protocol. We give a systematization of knowledge for how such a permutation correlation can be derandomized

Oblivious Pseudorandom Functions (OPRFs)

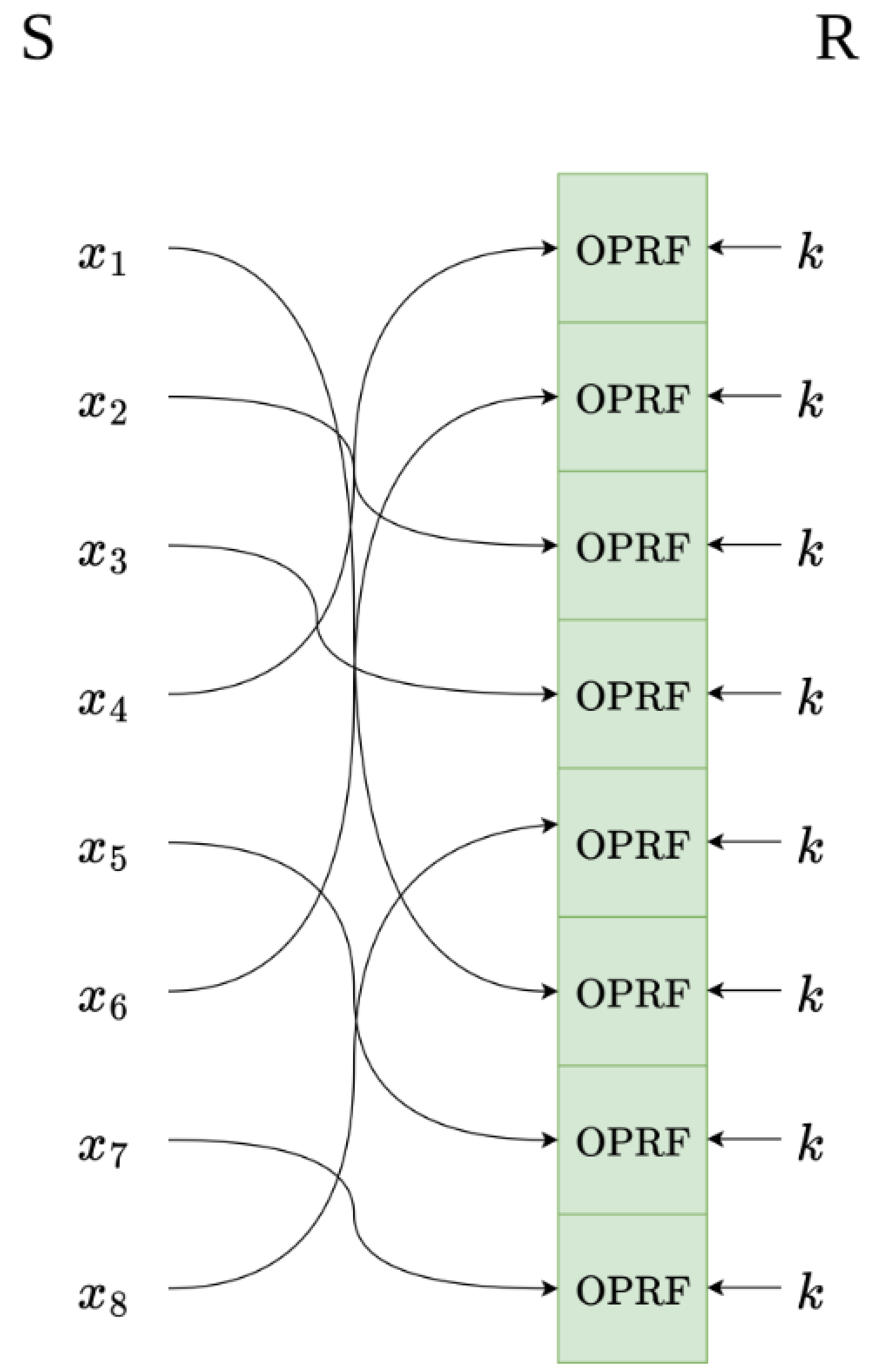
A core building block of each construction is an OPRF with secret-shared output.

- Sender provides an input x
- Receiver provides a key k
- Instantiated with the alternating-moduli family of PRFs
- Outputs a secret sharing



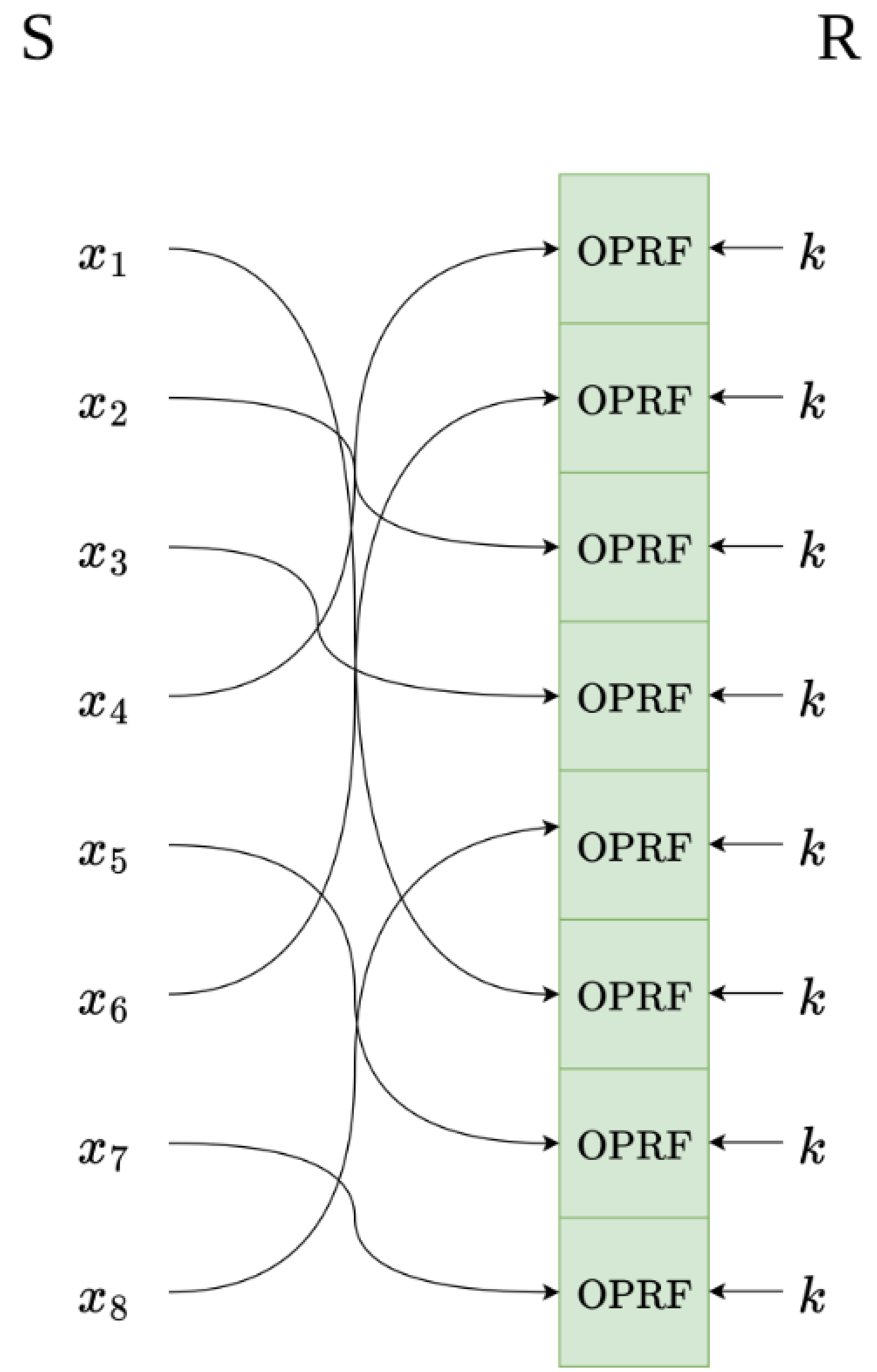
Primary Construction

- Parties agree on a random vector $\mathbf{x}$ by computing $x_i = H(i)$
- S samples $\pi \leftarrow S_n$
- S permutes $\mathbf{x}$ by π to set OPRF inputs
- R samples a random PRF key k
- R sets $A_i = F_k(x_i)$ in plaintext
- Parties set B and C as the OPRF outputs



Analysis of the Primary Construction

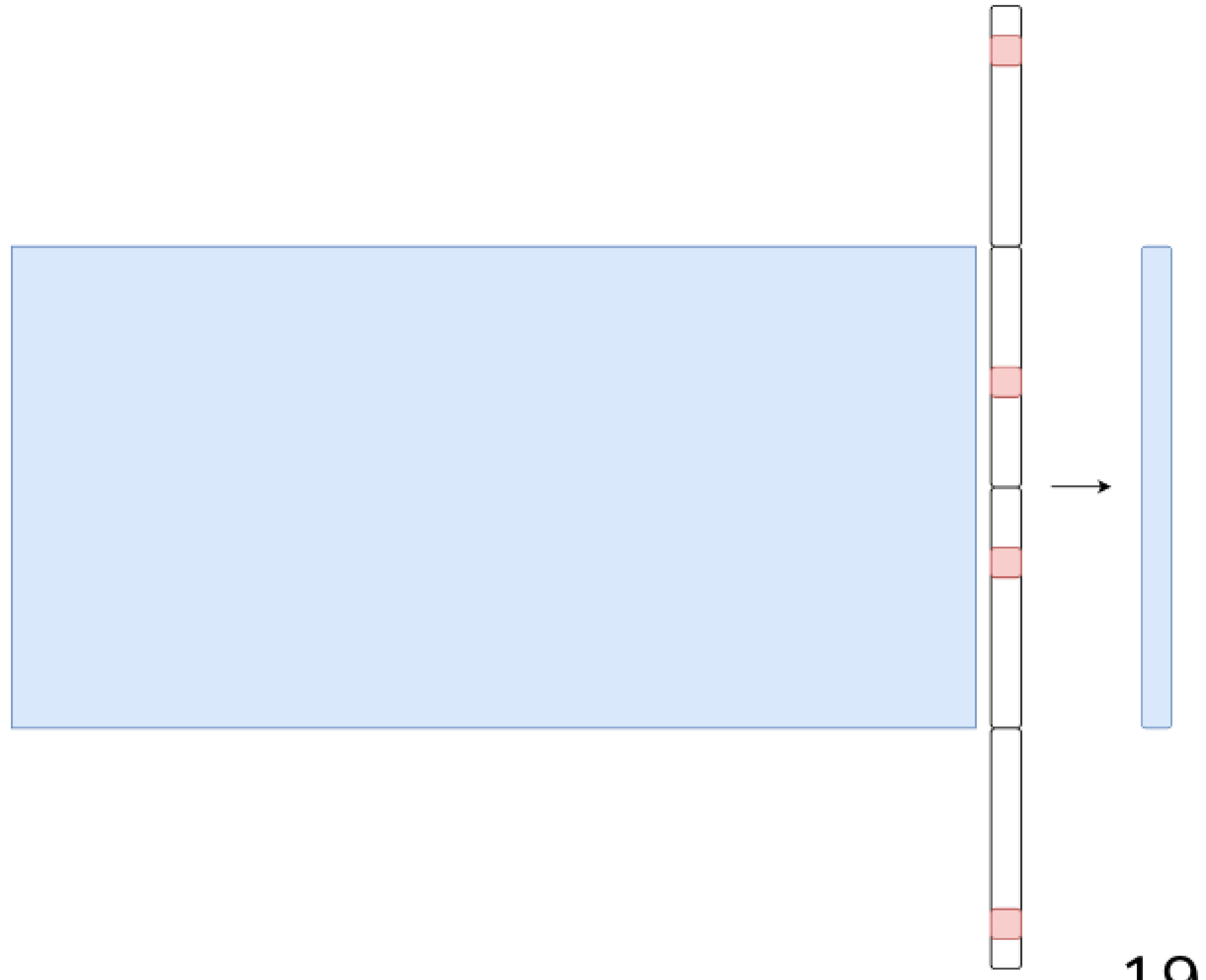
- n calls to the OPRF
- $O(n\ell)$ total communication
- Most efficient construction to date in practice
- ≈ 4s per million elements



Sublinear Protocol

Treat the output of the primary construction as a vector of seeds to pseudorandom correlation generators.

- Let t be some sparsity parameter
- Each $\log(2\ell/t)$ bits encodes a unit vector of length $2\ell/t$
- Compute each unit vector using a distributed point function (DPF)
- Concatenate t unit vectors to obtain a t-sparse vector of length 2ℓ
- Compress into a pseudorandom vector of length ℓ
  - Reduction to syndrome decoding assumption



Analysis of the Sublinear Protocol

- $O(n \log \ell)$ total communication
- Efficient only for very long ℓ
- Crossover point is $\ell \approx 4000$
- Application: graph algorithms
  - Requires thousands of bits of permutation correlations

Open Problems

Goal: Extend the line of work on PCGs into the domain of oblivious shuffling.

We want to do to oblivious shuffling what pseudorandom correlation generators did to oblivious transfer.

Concrete technical problems:

- Sublinearity in ℓ (in practice)
- Sublinearity in p
- Sublinearity in n

Open: Sublinearity in ℓ

Protocols exist for generating permutation correlations with sublinear communication in ℓ.

- No implementation of the Peceny et al. construction (until now!)
- Improve parameters to get practical efficiency

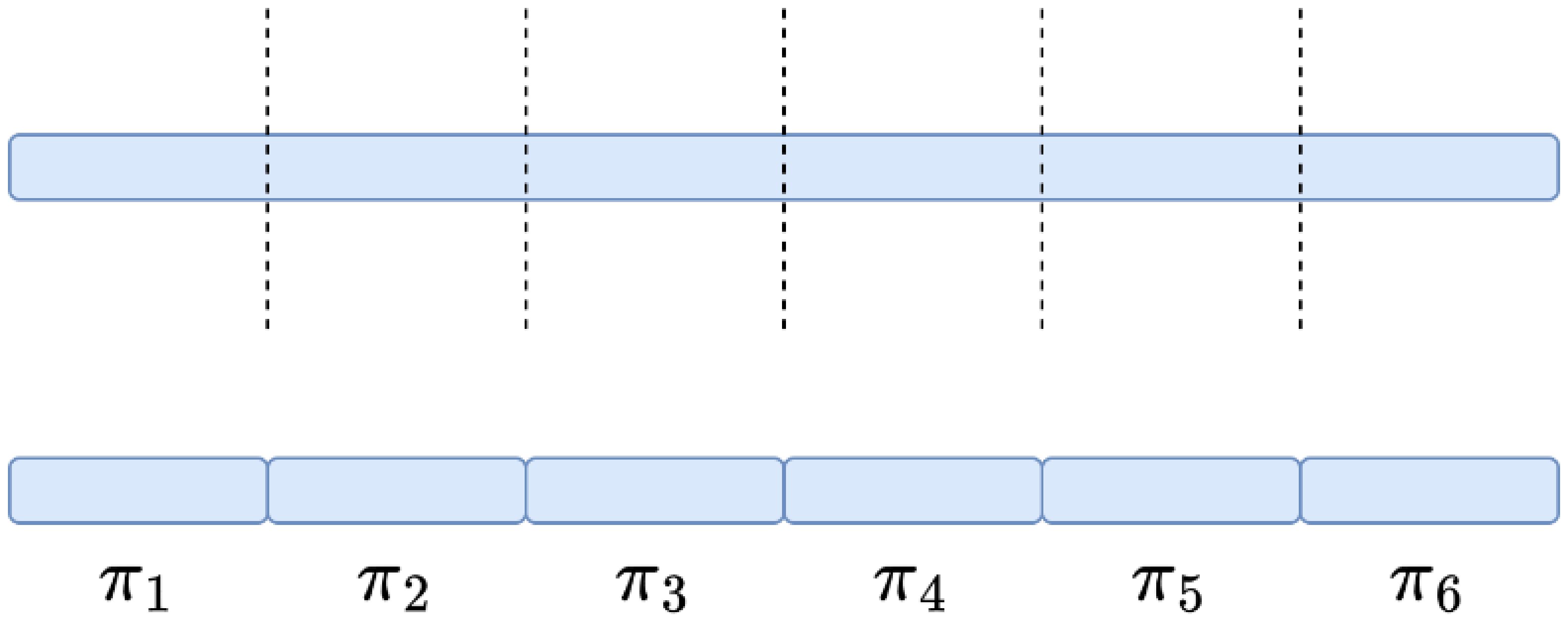
Ongoing work: graph algorithms

Open: Sublinearity in p

Can we "chop up" a correlation with large ℓ into many correlations with small ℓ?

- Very challenging to achieve independence
- Can define an ideal functionality that captures the intended behavior

Application: relational analytics



Open: Sublinearity in n

The holy grail of this research direction.

- Make shuffling an efficient building block for *any* workload

Preliminary ideas:

- Generate many small correlations, then "stitch them together"
- Construct from a protocol with sublinearity in p
- Generate small correlations directly by materializing the permutation matrix

Roadmap

~~1. Oblivious Shuffle Preprocessing~~

2. Scaling with Many Parties

Scaling with Many Parties

Motivation

Challenge: Suppose a secure computation involves thousands of parties.

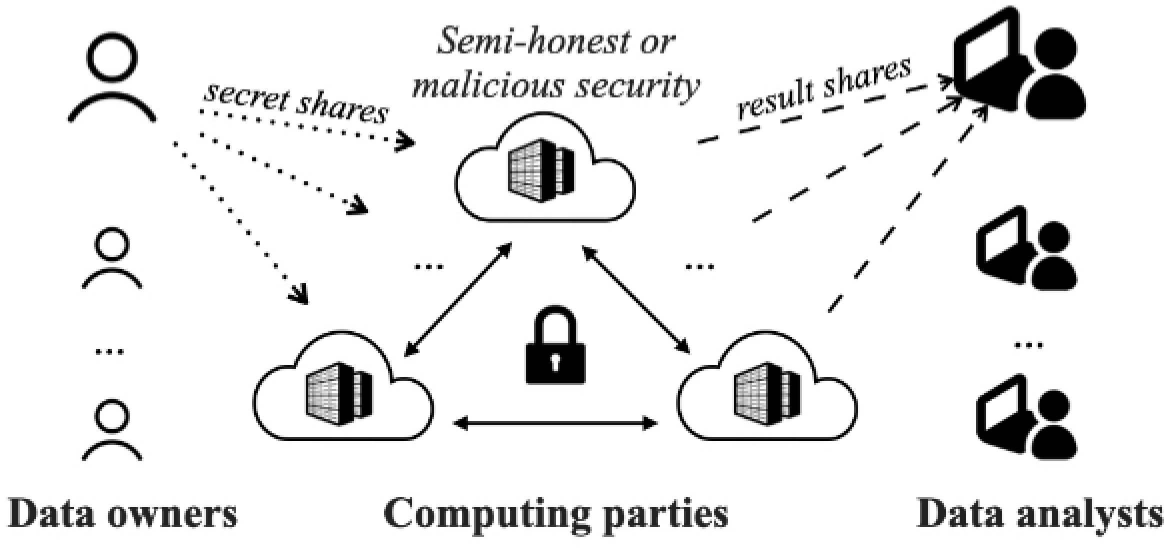
N Party Computation

- Large body of theoretical work
- Prohibitive practical challenges

The Outsourced Setting

- Practical and efficient
- Requires stronger trust assumptions

Goal: Scale *practical* MPC to thousands of parties.



Two Lines of Work

Secure Aggregation

- Scales to thousands of parties
- Restricted to very simple functions

MPC with a Star Topology

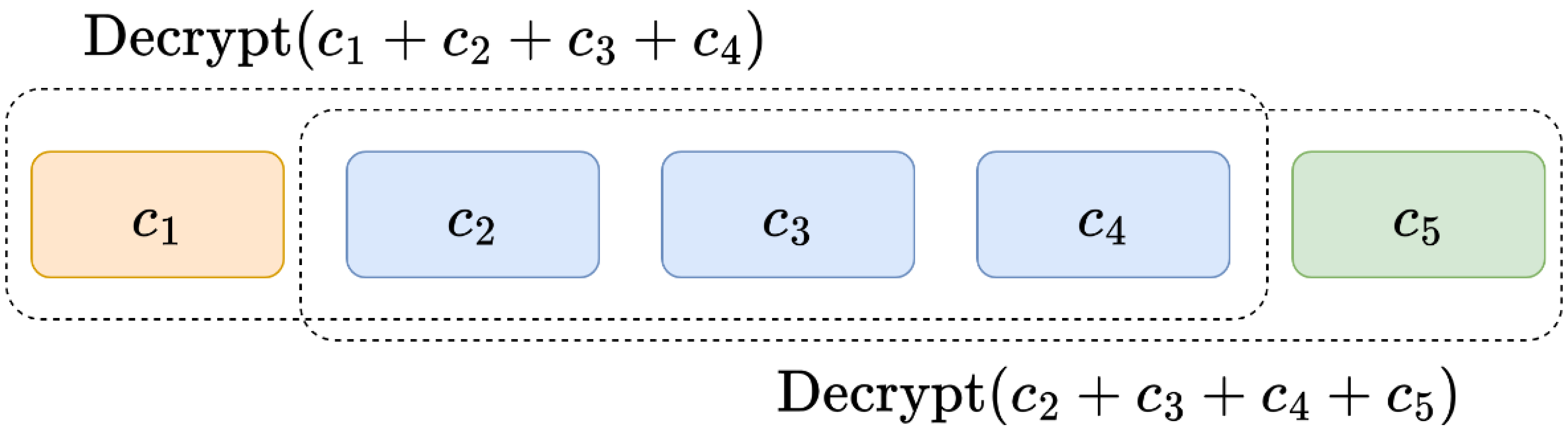
- Scales to dozens of parties
- Fully general

Secure Aggregation

Many parties, very simple computations (sum).

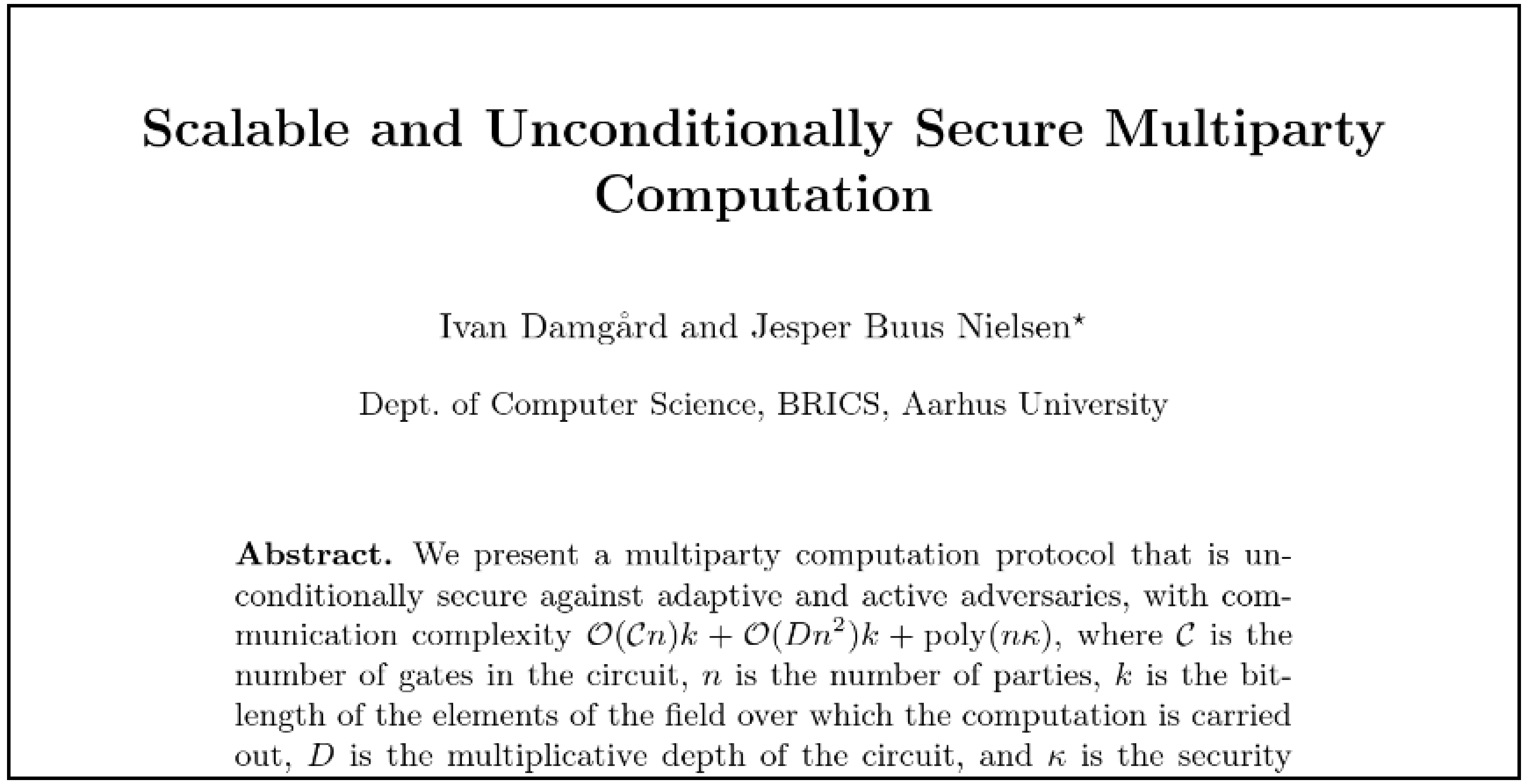
- Ephemeral clients send encryptions to a powerful untrusted central server
- Design goal is non-interactivity
- Naive approach is threshold homomorphic encryption
  - Vulnerable to "residual attack"

The emerging paradigm uses a small committee to atomically decrypt (OPA, Willow, TACITA).



A Star Topology

- Damgard-Nielsen, 2007
- Central untrusted "king" party
- $O(n)$ communication per gate
- Based on Shamir secret sharing
- Requires an honest majority

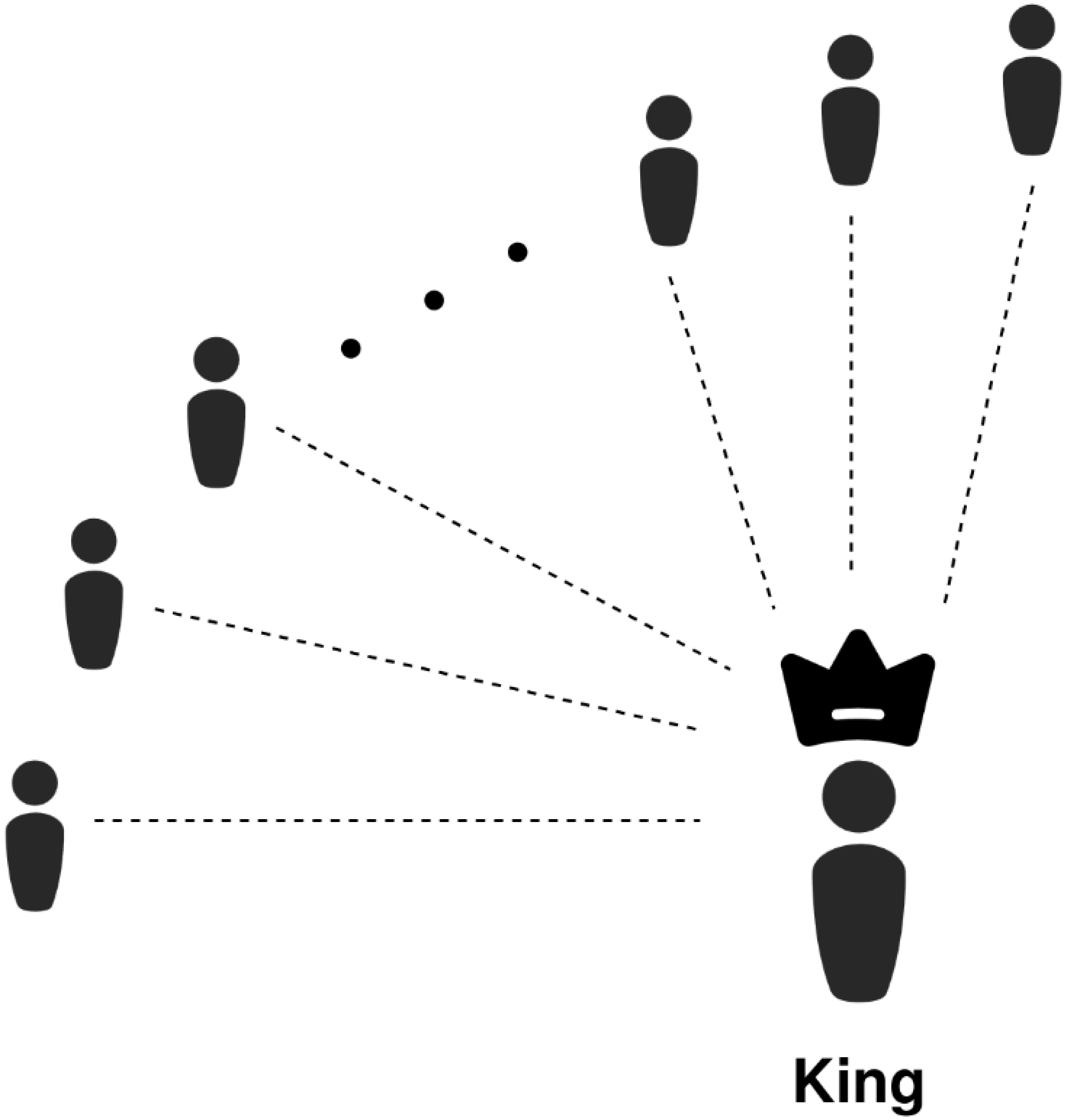


Scalable and Unconditionally Secure Multiparty Computation

Ivan Damgård and Jesper Buus Nielsen*

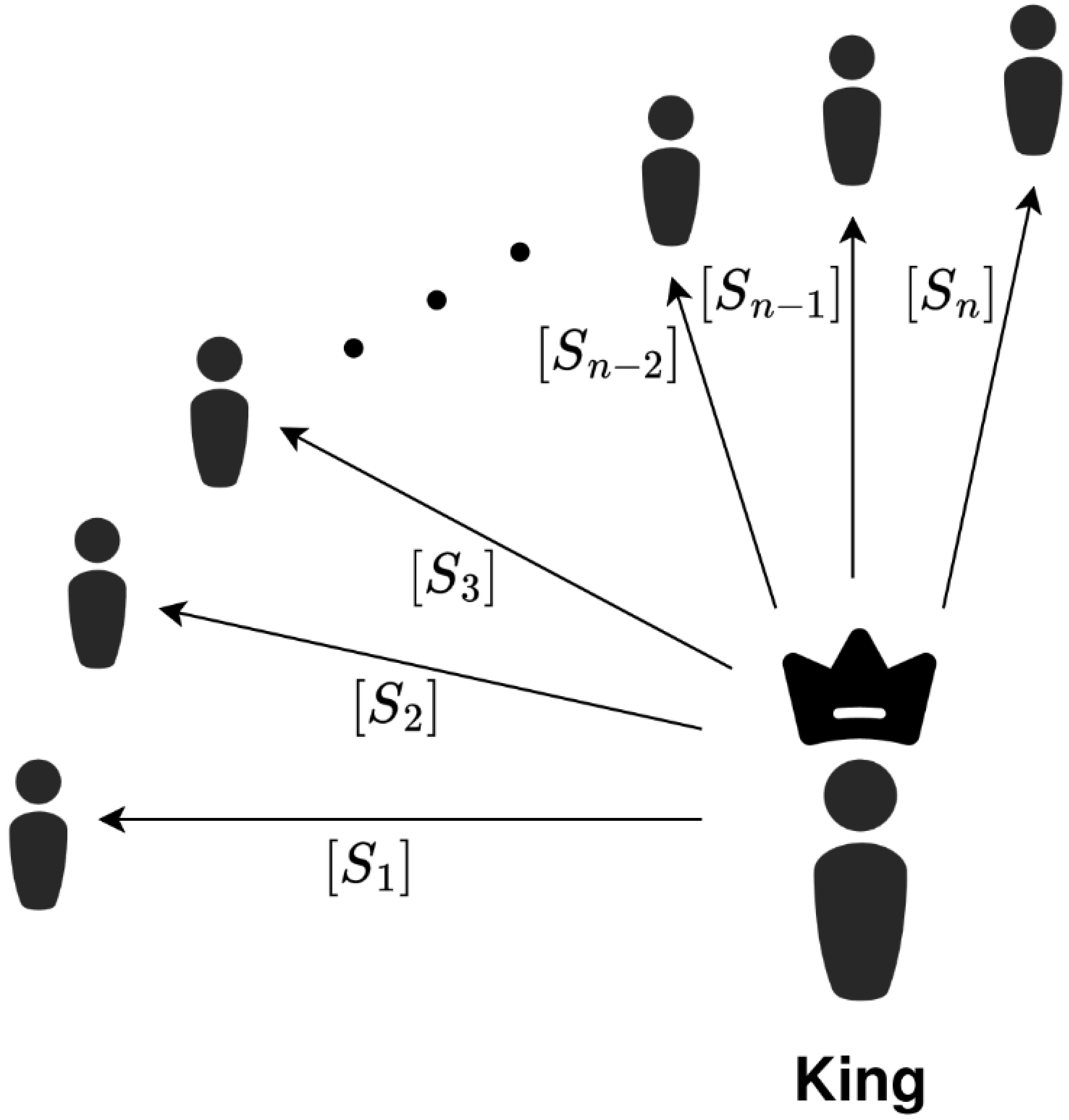
Dept. of Computer Science, BRICS, Aarhus University

Abstract. We present a multiparty computation protocol that is unconditionally secure against adaptive and active adversaries, with communication complexity $\mathcal{O}(\mathcal{C}n)k + \mathcal{O}(Dn^2)k + \text{poly}(n\kappa)$, where $\mathcal{C}$ is the number of gates in the circuit, n is the number of parties, k is the bit-length of the elements of the field over which the computation is carried out, D is the multiplicative depth of the circuit, and κ is the security



DN07 Multiplication

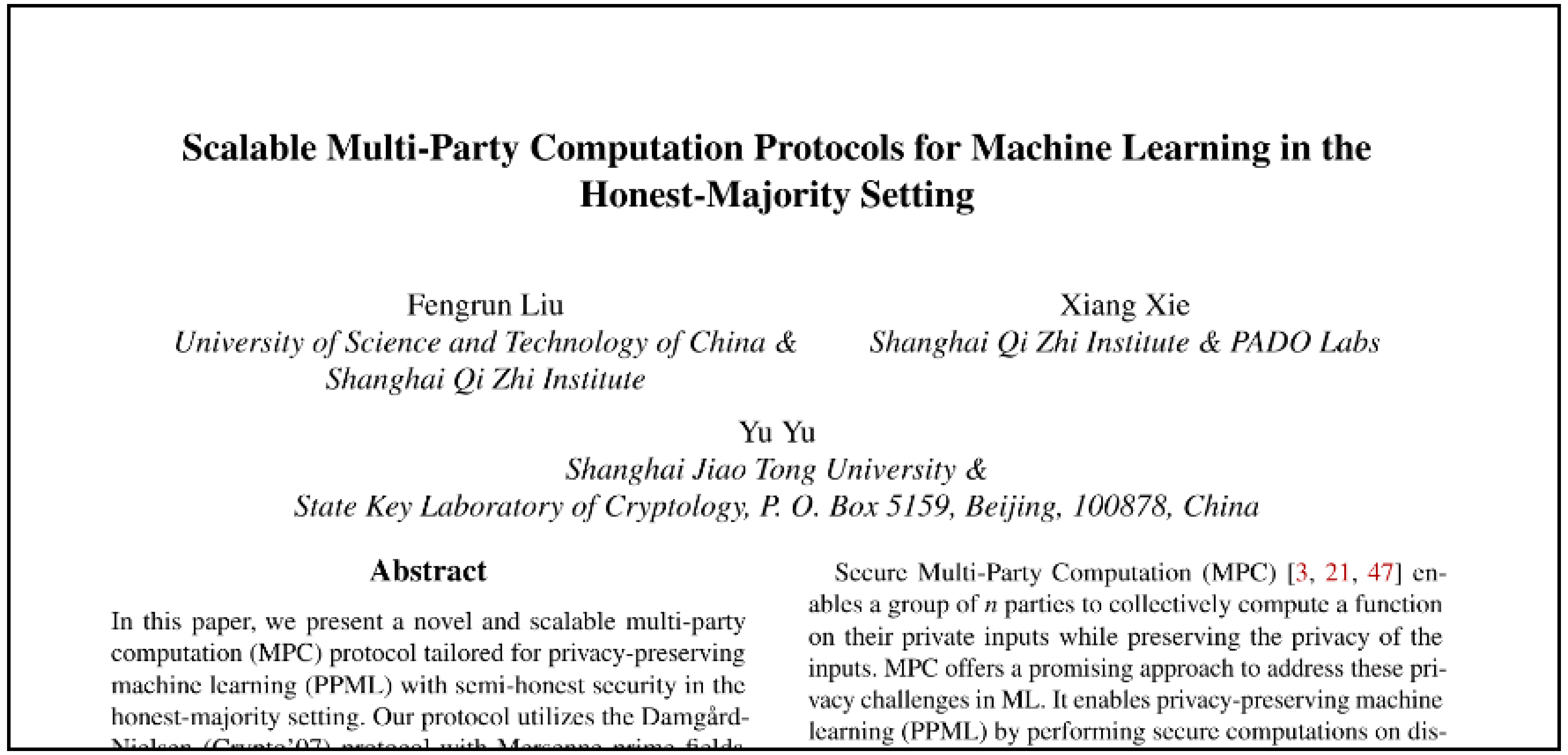
- $[x]$ denotes a degree-t Shamir sharing
- $[[x]]$ denotes a degree-$2t$ Shamir sharing
- Local multiplication converts t sharing to $2t$ sharing
- We need to perform a degree reduction
- Multiplication consumes a double sharing $([r], [[r]])$



Extending to More Complex Functions

Liu et al. (Usenix Security 2024)

- Based on the DN07 protocol
- Uses the properties of Mersenne prime fields
- Bitwise operations on arithmetic secret shares
- Targets machine learning inference
- Optimizes for round complexity
- Shows results with up to 63 parties



Scalable Multi-Party Computation Protocols for Machine Learning in the Honest-Majority Setting

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Xiang Xie
Shanghai Qi Zhi Institute & PADO Labs

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Abstract

In this paper, we present a novel and scalable multi-party computation (MPC) protocol tailored for privacy-preserving machine learning (PPML) with semi-honest security in the honest-majority setting. Our protocol utilizes the Damgård-

Secure Multi-Party Computation (MPC) [3, 21, 47] enables a group of n parties to collectively compute a function on their private inputs while preserving the privacy of the inputs. MPC offers a promising approach to address these privacy challenges in ML. It enables privacy-preserving machine learning (PPML) by performing secure computations on dis-

Liu et al. Truncation Protocol

- Division by a public power of two

Almost all truncation protocols either

- Require an arithmetic-to-binary conversion
- Require "slack bits" to prevent a "catastrophic" error

The Liu et al. protocol is both constant-round and requires only 1 bit of slack.

- In a Mersenne prime field, the catastrophic error is predictable and can be corrected for

The paper simultaneously minimizes total work and makes base operations constant-round.

Liu et al. in Practice

LAN Setting

- Bottlenecked by local computation on the king party
- King needs to perform polynomial interpolation for each gate

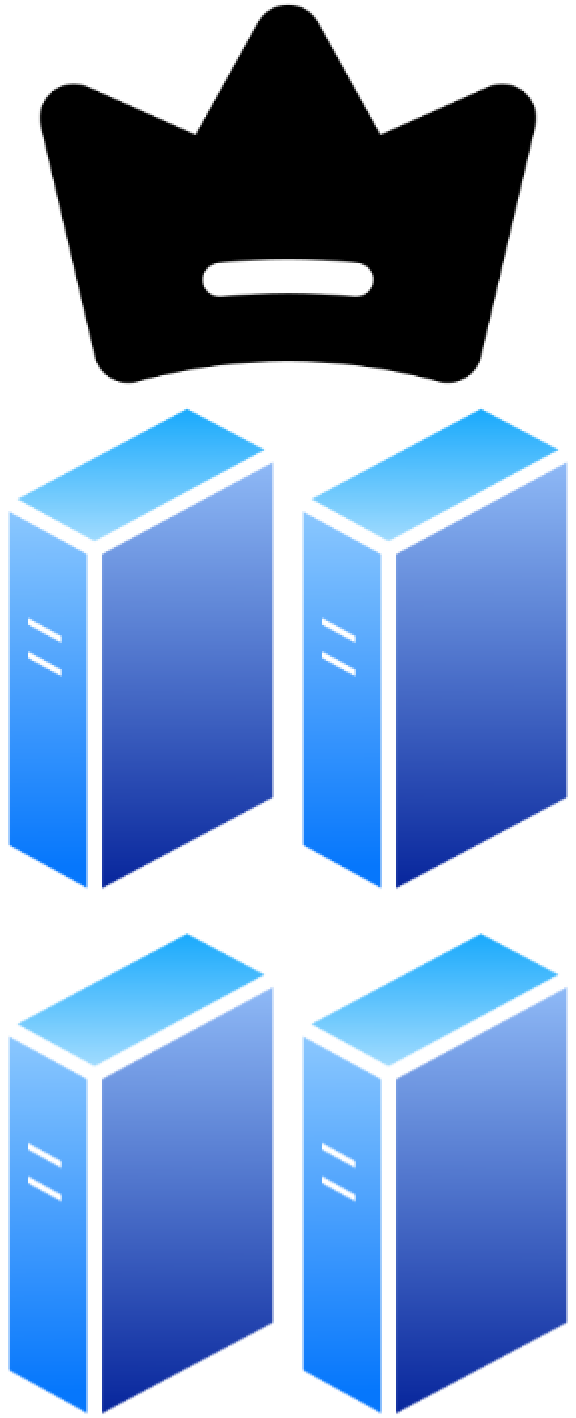
WAN Setting

- Latency constrained
- Uses 5% of available bandwidth

Future: Scaling the King Party

Proposal: Lean into the king as the bottleneck.

- Powerful central server, weak ephemeral clients
- One logical king, multiple servers
- GPU acceleration

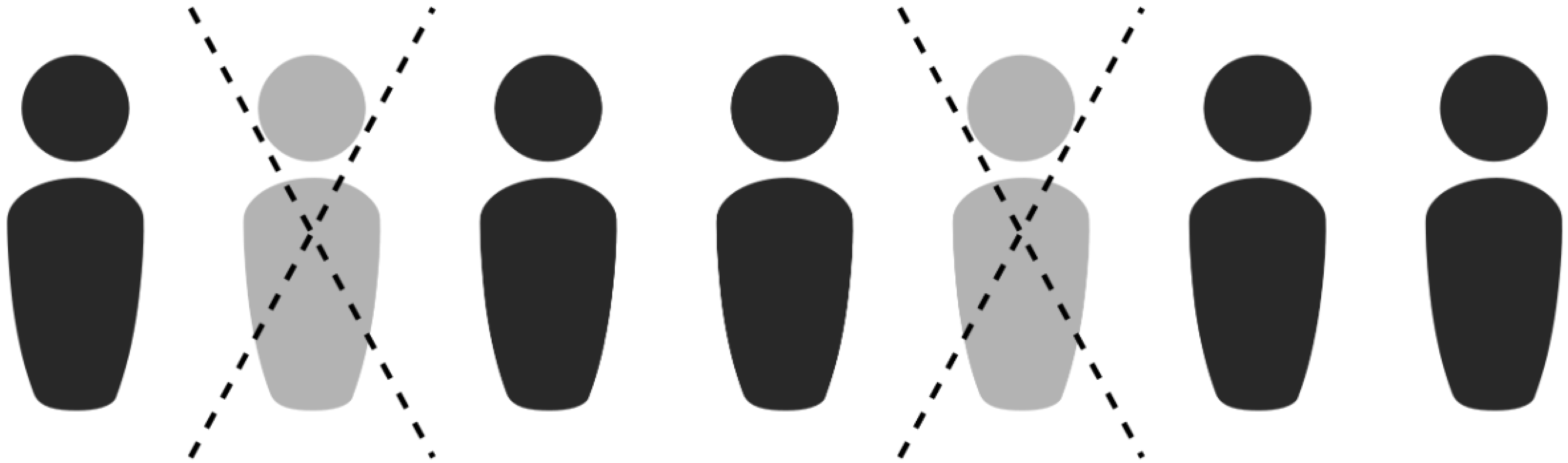


Future: Tolerating Dropouts

Problem: Slowest connection becomes the bottleneck.

Proposal: Embrace the threshold scheme to tolerate dropouts.

- Subset of Fluid MPC
- Semi-honest security is straightforward
- Malicious security is more complex



Summary

Vision: Efficient secure computation involving millions of parties for complex functions.

- Oblivious shuffle as a way to increase the expressivity of MPC
  - Incorporating pseudorandom correlation generators

- Approaches to scale moderately complex workloads to thousands of parties
  - Embracing the star topology
  - Tolerating dropouts

Thank You!